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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Lets go
Project / Solution Name	Milestone2Cash Proof-to-Cash
Name (Leader)	Jiazhong Yang
Email & Phone (Leader)	yjzae41475@gmail.com Phone: +60 173072181
University Affiliation(s)	Monash University Malaysia; Universiti Malaya
Live Prototype URL	https://milestone-2-cash-proof-to-cash.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/jake2ace/SDC-Tracker Public repository; visibility verified on 25 October 2026.

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Milestone2Cash connects a React interface to a Node.js TypeScript HTTP API and SQLite database. Project completion creates an invoice draft for Owner approval. Finance reviews reminder drafts and customer replies. Tesseract.js reads uploaded synthetic receipts locally; a restricted text projection is sent to the DeepSeek API for structured extraction. The submitted code includes schema, evidence, confidence and semantic checks for uncertain, negative, conditional or redacted replies. A deterministic matcher compares receipt evidence with imported bank CSV credits. Finance approval rechecks the current proposal, permissions, remaining amount and duplicate allocation before a transaction records the payment and audit event. Partial payment reduces only the outstanding balance. A separately confirmed promise changes an expected receipt date, not actual cash. The cash screen calculates 30-day on-time and delayed scenarios. Replit serves the built frontend and API at one public address. The published SQLite filesystem is a resettable demo, not durable financial storage. Bank, messaging, tax and Orbyte integrations are simulated.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node.js 24.x; built-in SQLite	HTTP API, transactions and persistent local demo records. Minimum Node 22.13.	Node.js licence; SQLite core public domain
TypeScript 5.9.3	Application and shared contract types; type checking	Apache-2.0
React / React DOM 19.3.1	Interactive frontend	MIT
Vite 8.3.5; React plugin 6.1.2	Frontend development and production build	MIT

Name & Version	Purpose in Project	License
Tesseract.js 7.0.0; eng / chi_sim data 1.0.0	Local OCR with pretrained English and Simplified Chinese data	Apache-2.0; data package metadata MIT; retain model notices
sharp 0.35.5	Image decoding, validation and synthetic test image generation	Apache-2.0; bundled libvips notices include LGPL
Playwright Test 1.64.1; Vitest 5.0.5	Browser integration and frontend tests	Apache-2.0 / MIT
Ajv 8.20.0	JSON Schema and fixture validation	MIT
@types/node 22.20.5; @types/react and react-dom 19.3.0	Development type definitions	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
DeepSeek chat-completions API https://api.deepseek.com	Structured receipt/reply extraction from restricted text; reminder draft generation	Team-provided API account; exact billing tier not recorded. Not sponsor-provided.
Other live business APIs: None	No live bank, payment, WhatsApp, MyInvois or Orbyte calls. Sending changes synthetic records only.	Not applicable

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Team workspace; public frontend and API hosting. Project and deployment links appear in the proof-of-use PDF.	Reserved VM, 0.5 vCPU / 2 GiB, Asia, as recorded in hosted verification.
GitHub	Source history, issues, pull requests and submission distribution	Public repository; visibility verified on 25 October 2026. Account plan not declared.
DeepSeek hosted API	Application inference	Provider-managed service; account-specific processing location and retention terms not recorded.
RunPod: not used	No RunPod GPU workload or endpoint was deployed	Not applicable; no RunPod resources used.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
deepseek-flash	DeepSeek hosted API	Pretrained hosted model; no team training or fine-tuning	Hosted API provider terms; no separate open-weight licence claimed
Tesseract eng / chi_sim trained data, packages 1.0.0	@tesseract.js-data	Pretrained OCR data, local inference; no team training	Bundled package/model notices; package metadata MIT
Matcher / forecast: not AI models	Team-authored deterministic code	Rules and scenario calculations; no trained prediction model	Project code: Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo companies, roles, projects, invoices and bank records	Repository data/seed/	Synthetic runtime fixtures	Project-created; Apache-2.0	N
40 receipt labels and generated images, four layouts	Repository data/eval/; scripts/generate-receipts.ts	Synthetic evaluation; separate inputs and gold labels	Project-created; Apache-2.0	N
40 English, Chinese and Malay replies	Repository data/eval/ replies inputs and gold labels	Synthetic evaluation	Project-created; Apache-2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
30 matching cases and expected results	Repository data/eval/ matcher inputs and gold labels	Synthetic evaluation	Project-created; Apache-2.0	N
4 forecast cases with expected daily rows	Repository data/eval/ forecast inputs and gold labels	Synthetic evaluation	Project-created; Apache-2.0	N
Partial/final receipts and final-payment bank CSV	Repository data/demo/	Synthetic demonstration; separate from frozen evaluation	Project-created; Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex; model: GPT-6 Astra	Backend/frontend code, UI/routing, automated tests, agent-operated online browser validation, research, documentation support	Major assistance
Anthropic Claude; model: Fable 5.1	Frontend development support and interface implementation	Major assistance; tool use confirmed by the team

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Validation includes TypeScript checks, 60 backend tests, 23 frontend tests and 13 contract/data groups. Separately, on 25 October 2026, Codex operated the public prototype through six browser workflows (F01-F06), all passing. Four UI-labelled AI outputs covered a reminder, two receipts and a promise; the receipts and promise displayed 95% confidence and needed no correction. Approval gates, partial/final balances, duplicate handling, company isolation and narrow-screen language/theme checks passed. These are agent-operated tests, not independent member or SME validation. Final member review and responsibility remain to be confirmed before submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Development workspace and hosting platform	Existing team project built and publicly hosted. Actual project/deployment evidence accompanies this declaration.
Alpha Red Solutions / Orbyte	Domain 2 challenge and platform context	Product modularity and integration assumptions; no live Orbyte API or production access claimed.
RunPod	None	Not used. Inference calls the DeepSeek hosted API; the team did not operate a RunPod GPU workload.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit hosts the public prototype and supports the development workspace. The Node.js, SQLite and React stack is portable. Model inference uses the team's existing DeepSeek API account. Orbyte interfaces are simulated. RunPod was not used; no model training or fine-tuning was performed. Replit project and deployment evidence accompanies this declaration.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype accepts only synthetic demonstration data. SQLite stores invoices, customer replies, uploaded original receipt bytes, OCR text, corrections, approvals and audit events. Tesseract runs locally. DeepSeek receives a restricted text projection; receipt names and references are tokenised and unrelated reply text is redacted. Original images and full bank CSVs are not sent to the model. Server-side environment variables hold API keys. Company-scoped role checks restrict reads and writes, including receipt images. Demo identity selection is simulated and is not production authentication. Data remains until reset or deletion, and redeployment can reset hosted data. No automatic retention schedule is implemented. Provider retention and processing location require account-specific confirmation before real use. No real customer or financial records should be uploaded.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

OCR can confuse reference characters: the frozen synthetic evaluation scored 35/40 references, not real-bank accuracy. Model outputs may be wrong, low-confidence, ungrounded or unavailable. Schema/evidence checks, negation and condition safeguards, and manual correction mitigate these risks. Finance reviews evidence before confirming promises or payments. A screenshot alone does not prove settlement. Current-proposal validation, database transactions and idempotency prevent stale or duplicate posting. Bank and messaging integrations are simulated, and demo roles provide no production authentication. Forecast dates are scenario assumptions, not learned predictions. The Replit deployment is a shared resettable demo; a recorded republish reset its ledger and uploads. No staff-time saving, independent team review or production persistence is claimed.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

Unchecked statements require captain review before submission. The GitHub repository is public, verified on 25 October 2026. Final member review remains to be completed. Source reviewed: c438678 (24 October 2026); live access was verified separately and does not identify the deployed commit.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Leader: Jiazhong Yang
Signature: Jiazhong Yang

Date signed: 25 October 2026